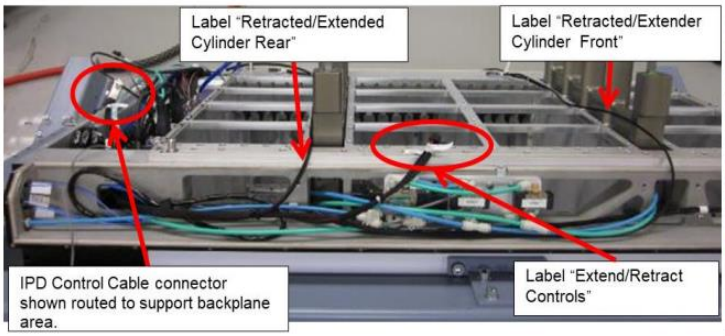
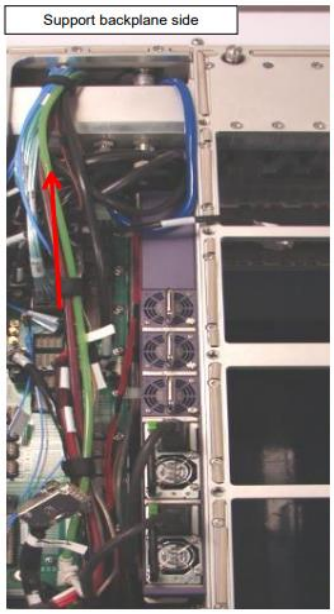
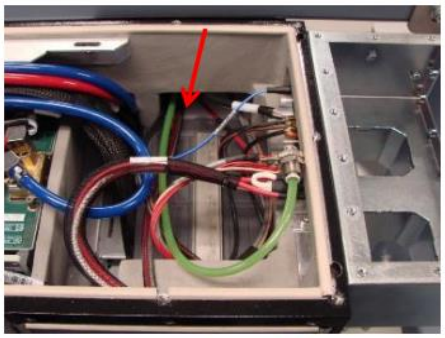
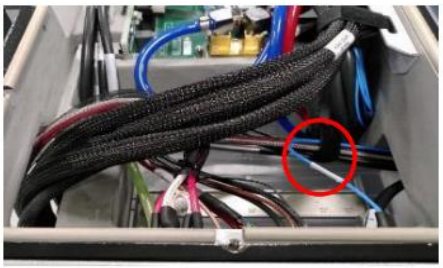
Build Overview:

1 **IPD Air Valve Install**
2 **NVRAM Cable Install**
3 **Support Board Cover Install**









CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by	
	1	New Issue	JOSEPH AMAD	**Prepared By:** JOSEPH R. AMAD, Manufacturing Engineer
				Checked By: IVAN NOWELL PANIQUE, Team Lead
				Approved By: SHERWIN ARIOSTE, Operations Manager

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	638-907-00	ELEC-MECH IPD VALVE SHUTL VER FOR SEMI TEST	1
2	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/ CONE WASHER. O.D. 11MM	2
3	813-929-01	HDW VELCRO STRP 5/8	9
4	639-820-00	CABLE IPD CONTROL W/ CONN	1
5	634-338-00	ALUMINUM COVER AIR HOSE BRACKET OR MISC ARTICLE	2
6	406-502-12	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	6
7	813-929-00	HDW VELCRO STRP 5/8	1
8	636-665-00	CABLE UF2_NVRAM_COMM 12 V W/ CONN	1
9	632-923-01	ALUMINUM COVER SBR BOTTOM PLATE BRACKET OR MISC ARTICLE	1
10	360-551-05	UOM:ea=96 inch SHIELD EMI GASKET, D-SHAPE, .197"HX.256"W, 96"L, UL94V-0 FLAM, PSA AND EXT LINR	1
11	406-512-12	SCRW CP SKT HD M5X.8P/12 MM SS W/NYLOK	13
12	653-411-00	PLASTIC LABEL 1-11 SLOT LABEL	1
13	653-411-01	PLASTIC LABEL 0-10 SLOT LABEL	1
14	653-411-02	PLASTIC LABEL 12-22 SLOT LABEL	1
15	653-411-03	PLASTIC LABEL 13-23 LABEL	1
16	640-211-00	PLASTIC ETHERNET SWITCH CABLE LABEL	1
17	656-411-00	PLASTIC, UF2 TH SBR LABEL, LABEL	1
18	638-517-04	PLASTIC CDU COOLANT RETURN LABEL	1
19	638-517-03	PLASTIC CDU COOLANT SUPPLY LABEL	1
20	635-536-00	CABLE UF2 SB24 AOC 1M 10 V TELECOM WITH CONN	2
21	710-048-10	TRANCEIVERS PIN SFP+ MULTI 10GBS HOT PLUG DUPLEX CN 850NM	4

METHOD:

Flow chart of operation steps

BPM preparation
↓
Preparation Tools
↓
Prepare Materials
↓
Complete Operation
↓
Inventory Materials
↓
Operation Check
↓
Inventory tool

MACHINE:

TOOL NAME	TORQUE VALUE
PH2 PHILLIPS TORQUE SCREWDRIVER	3.3 N.m
17MM SOCKET WRENCH	3.3 N.m
T25 TORX WRENCH	3.3 N.m
4MM HEX BIT TORQUE WRENCH	3.3 N.m
RULER	NA

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	638-907-00	ELEC-MECH IPD VALVE SHUTL VER FOR SEMI TEST	1
2	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/ CONE	2

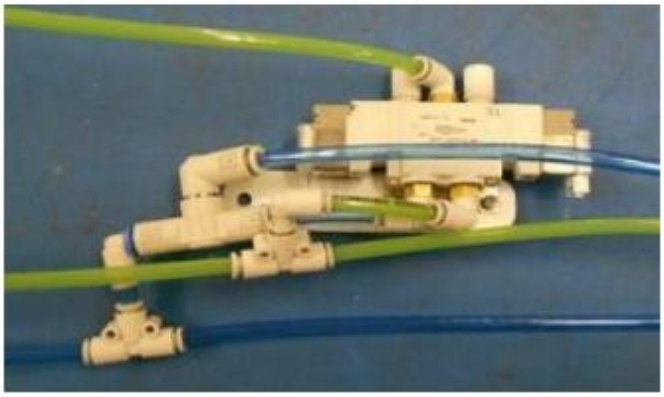
MACHINE:

PH2 Phillips torque screwdriver 3.3 N·m.

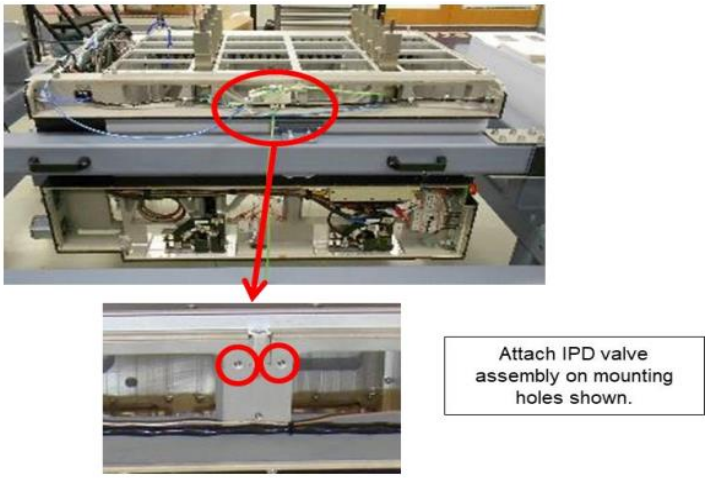
METHOD:

Operation steps

1.1 Take one IPD solenoid valve (638-907-00).



1.2 Rotate the T/H direction as shown, and install an IPD solenoid valve (638-907-00) on the T/H frame,
Fix the IPD solenoid valve with 2 M5 screws (472-426-12). The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

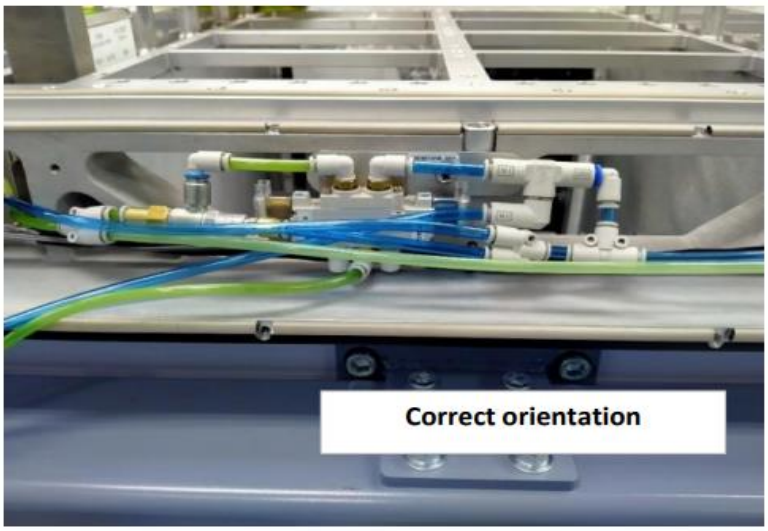
N/A

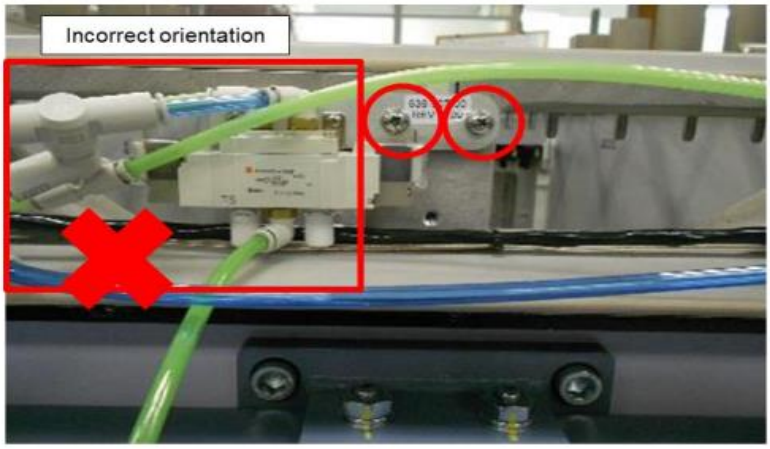
MACHINE:

N/A

METHOD:

1.3 Ensure that the T-shaped pipe joint is placed on the right side of the solenoid valve as shown in the figure.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

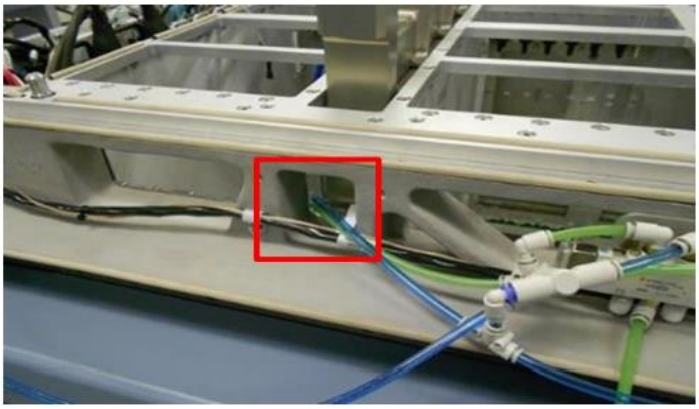
N/A

MACHINE:

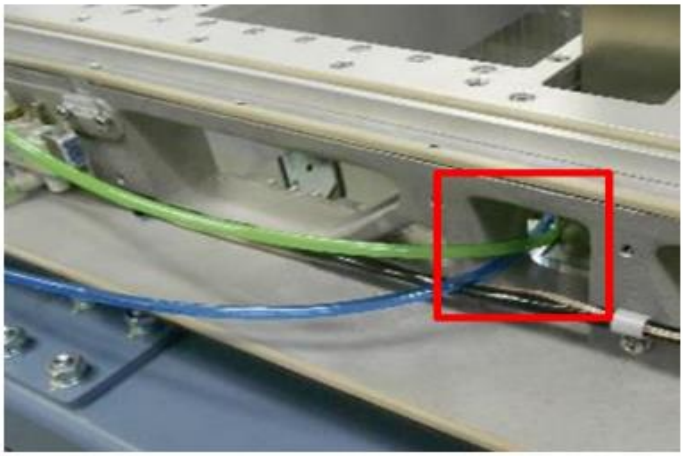
N/A

METHOD:

1.4 Insert a group of air pipes (blue, green) into the opening slot on the left side of the solenoid valve as shown, and enter the T/H frame.



1.5 Insert another set of trachea (blue, green) into the open slot on the right side of the solenoid valve as shown, and enter the T/H frame.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

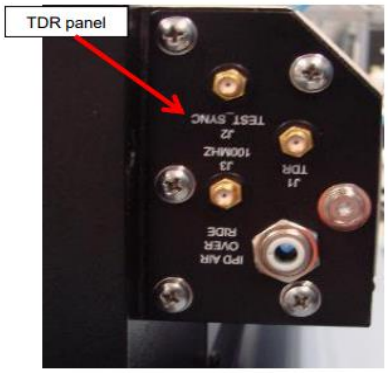
MACHINE:

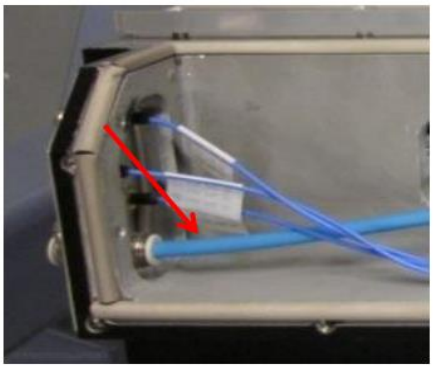
N/A

METHOD:

1.6 Remove the nut at the front of the blue external air pipe joint as shown in the figure, and adjust the distance of the nut behind the air pipe joint to 10mm away from the pipe joint.

1.7 Insert the blue gas tube into the TDR panel as shown in the figure, and install the nut.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

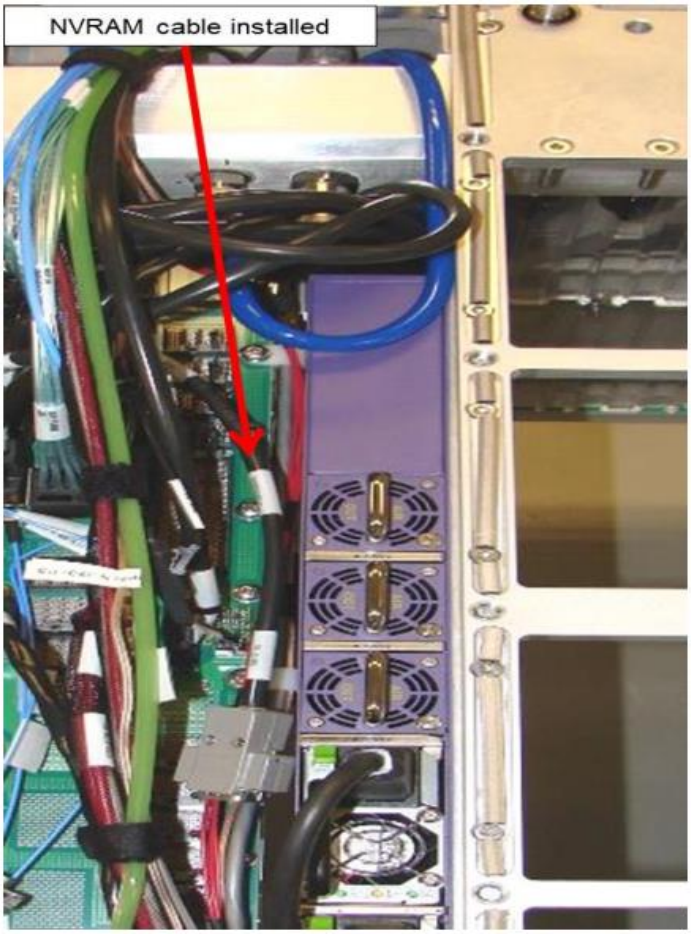
ITEM	PART NUMBER	DESCRIPTION	QTY
10	636-665-00	CABLE UF2_NVRAM_COMM 12 V W/ CONN	1

MACHINE:

N/A

METHOD:

7.1 Insert the connector of the NVRAM cable (636-665-00) labeled "Support Plane J38" into the J38 interface on the backplane PCBA as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

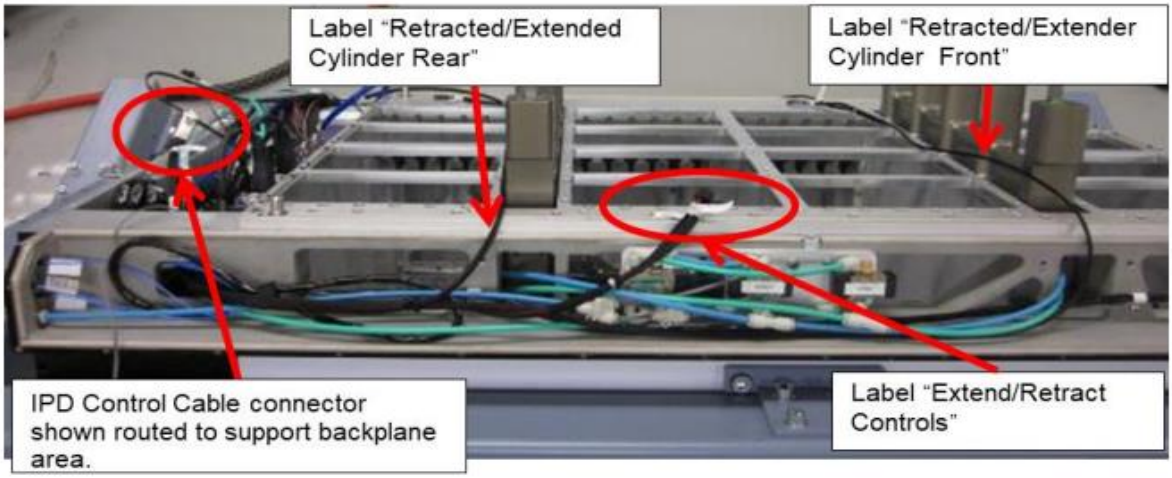
ITEM	PART NUMBER	DESCRIPTION	QTY
4	639-820-00	CABLE IPD CONTROL W/ CONN	1

MACHINE:

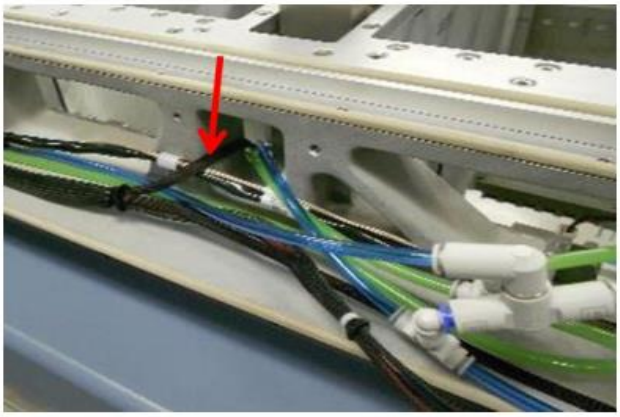
N/A

METHOD:

2.1 Take the IPD control cable (639-820-00) as shown in the figure, pass one end of the large cable connector through the T/H opening slot and place it in the backplane area.
(This connector will connect the PTL LCL board in subsequent assembly.)
2.2 Place the other end of the wire on the T/H as shown in the wire label as shown.



2.3 Insert the end of the line label "Retracted/Extended Cylinder Rear" into the T/H square slot on the left side of the solenoid valve as shown in the figure and into the T/H frame.



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

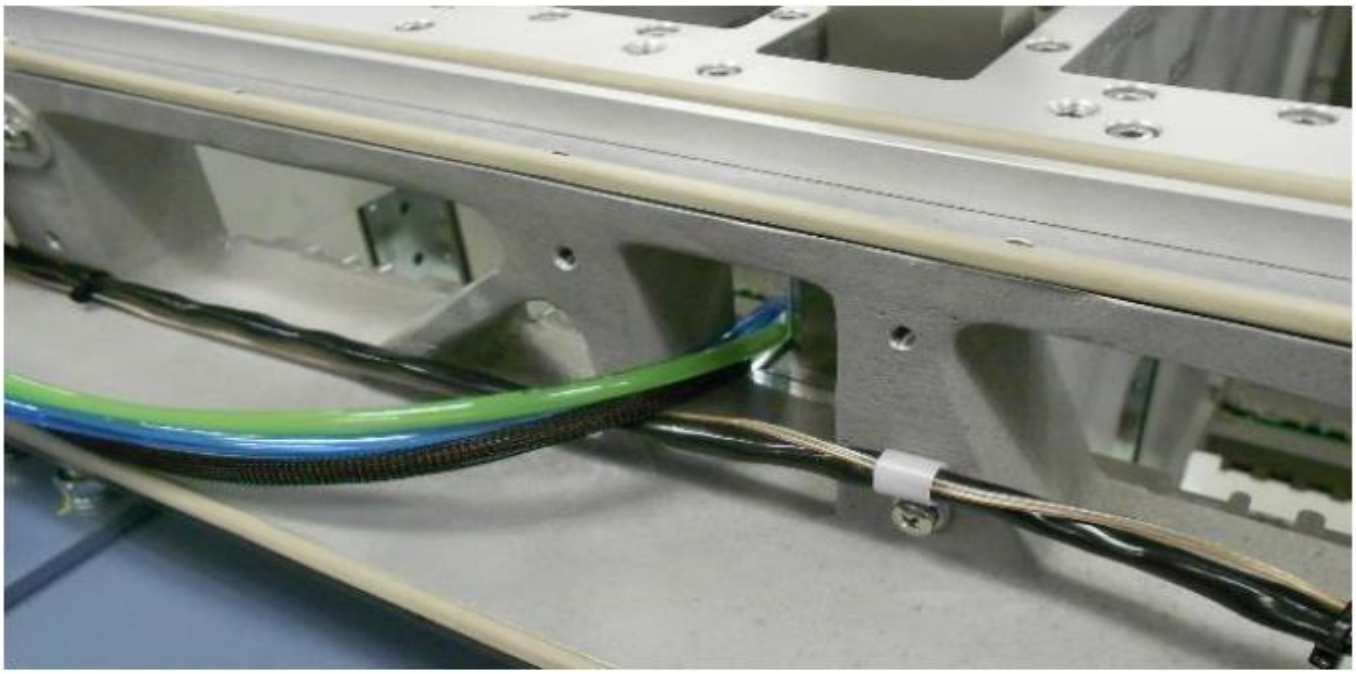
N/A

MACHINE:

N/A

METHOD:

2.4 Insert the end of the line label "Retracted/Extended Cylinder Front" into the T/H square slot on the right side of the solenoid valve as shown in the figure, and enter the T/H frame.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

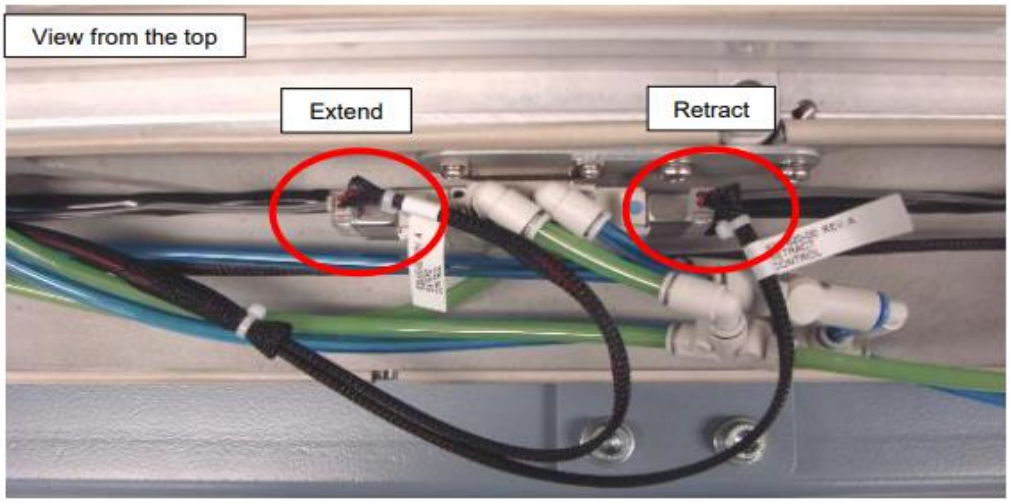
N/A

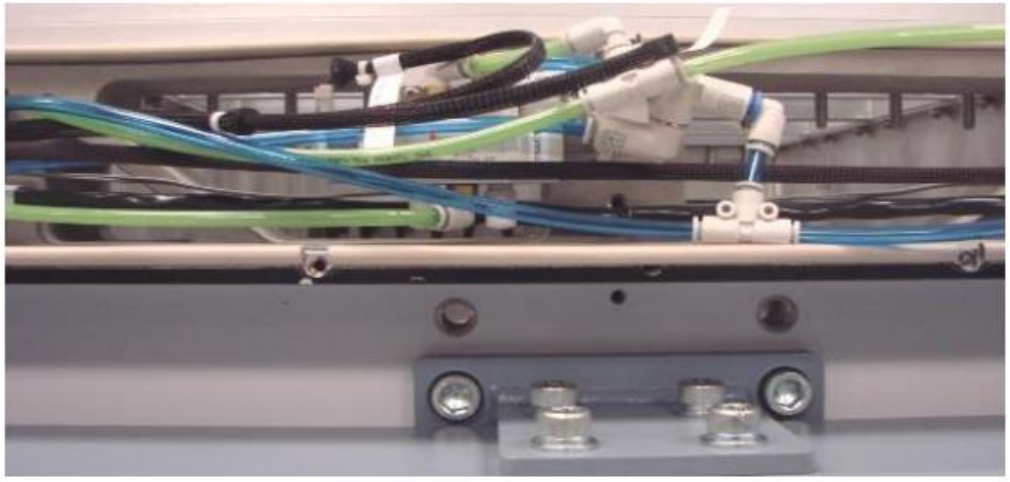
MACHINE:

N/A

METHOD:

2.5 Insert the connector of the line label "Extend/Retract" into the solenoid valve "Extend/Retract" as shown in the figure.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

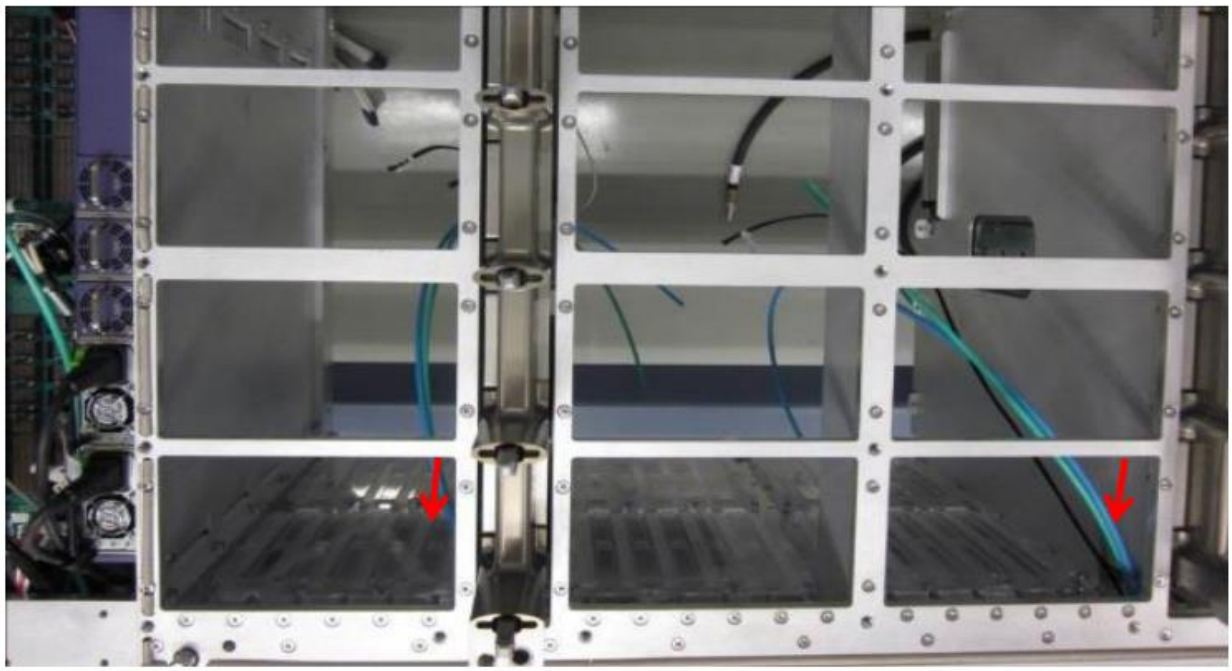
MACHINE:

N/A

METHOD:

2.6 As shown in the figure, the IPD line "Retracted/Extended Cylinder Rear" connector and "Retracted/Extender Cylinder Front"

IPD Retracted/Extended Cylinder Rear and Retracted/Extender Cylinder Front shown inside the card cage when viewing from DUT UP position of test head.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

N/A

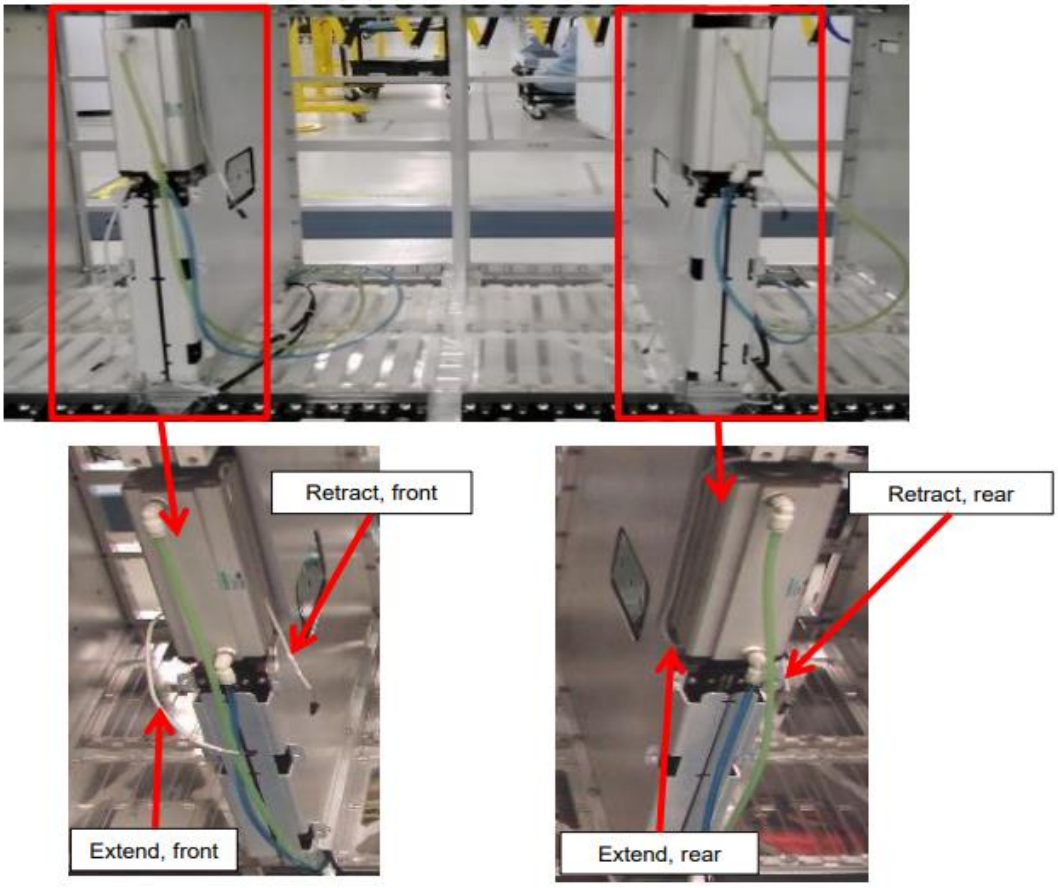
METHOD:

3.1 Remove the cable tie of the cylinder as shown in the figure, and place the "Retract/Extend" " signal line in the following directions:

a. "Retract" signal line is placed on the right side of the cylinder.

b. The "Extend" signal line is placed on the left side of the cylinder.

3.2 Insert the blue and green air pipes into the cylinder as shown in the figure, and adjust the elbow joint on the cylinder to the vertical direction.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

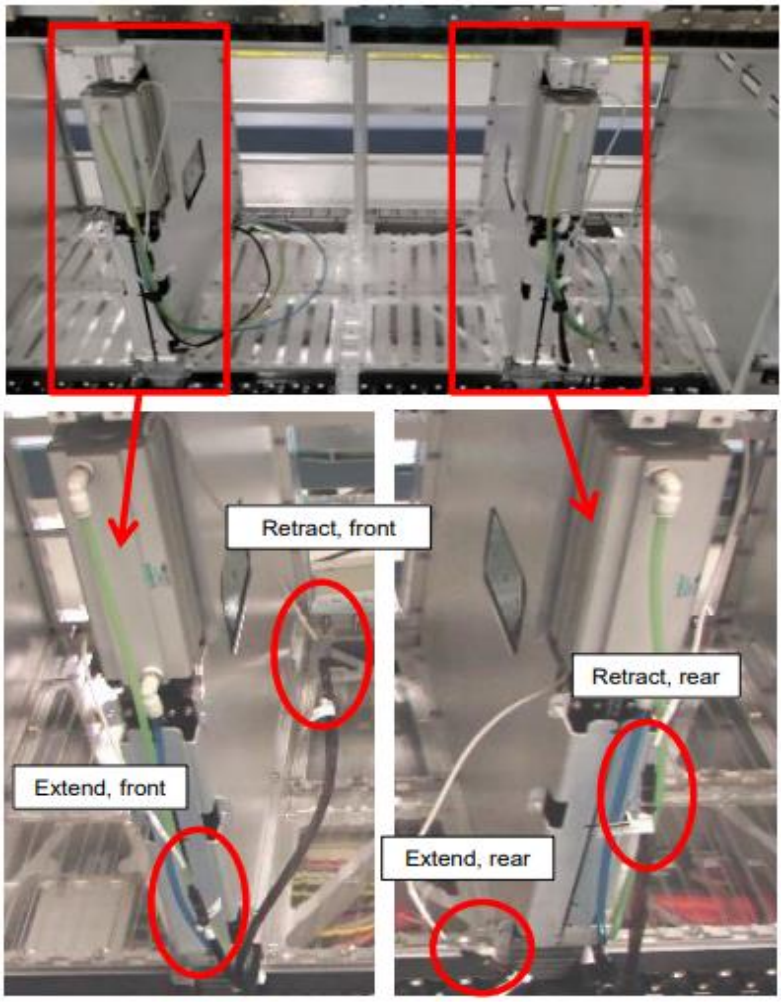
N/A

MACHINE:

N/A

METHOD:

3.3 As shown in the figure, connect the white line of the cylinder according to the label of the IPD control line. There are 4 positions in total.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
5	634-338-00	ALUMINUM COVER AIR HOSE BRACKET OR MISC ARTICLE	2
6	813-929-02	SCRW PAN HEAD TORX M5X.8P/12MM SS W/NYLOK	6
7	813-929-03	HDW VELCRO STRP 5/8	2

MACHINE:

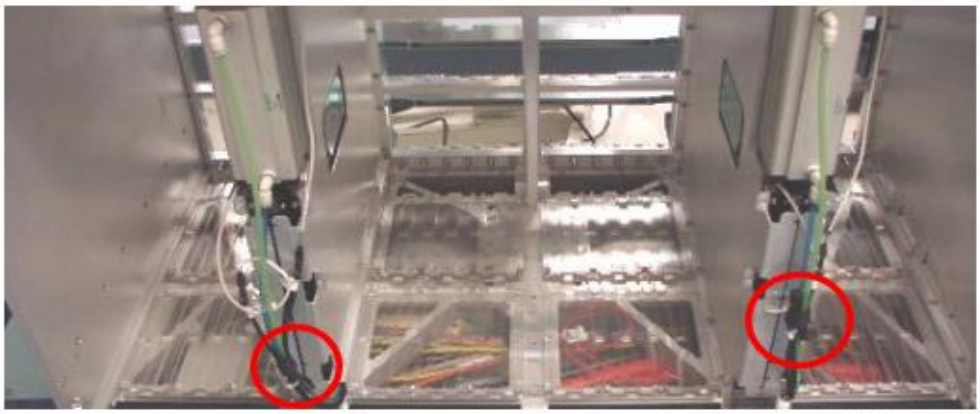
T25 Torx wrench 3.3 N·m

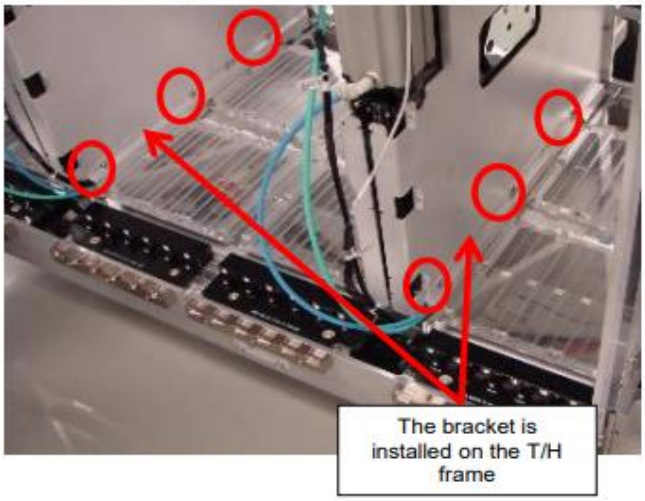
METHOD:

4.1 Take 2 IPD brackets (634-338-00), put the blue trachea, green trachea and control wire into the bracket.

4.2 Use 6 M5 screws (406-502-12) to fix the bracket on the T/H frame. The tool is a T25 Torx wrench and the torque value is 3.3 N·m.

4.3 Use 2 cable ties (813-929-01) to tie the trachea and the control wire together.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
8	634-338-00	HDW VELCRO STRP 5/8	2

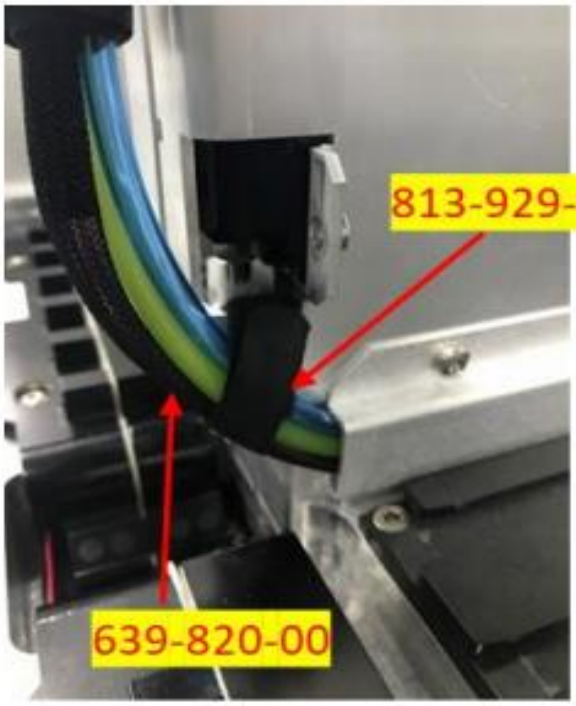
MACHINE:

N/A

METHOD:

4.4 Use another cable tie (813-929-01) to bind the air pipe and control wire together at the position shown in the figure, and keep the cable and air pipe away from the metal edge.

Repeat above step for both FRONT IPD and REAR IPD.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

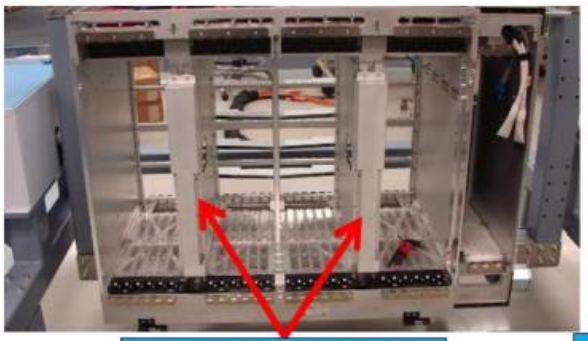
MACHINE:

4mm hex bit torque wrench 3.3 N·m

METHOD:

5.1 Reinstall the two IPD covers (636-889-00) as shown in the figure, and fix the cover with the 20 20 M5 screws removed before, The tool is a 4mm hex bit torque wrench with a torque value of 3.3 N·m.

Note: Make sure that when installing the cover, the air pipe and IPD control line are all inside and are not jammed by the cover!



IPD covers installed

Correct installation
639-820-00 cable free from squeeze

Incoorect installation-
639-820-00 cable squeeze

Incorrect installation
639-820-00 Cable squeeze

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
9	813-929-00	HDW VELCRO STRP 5/8	1

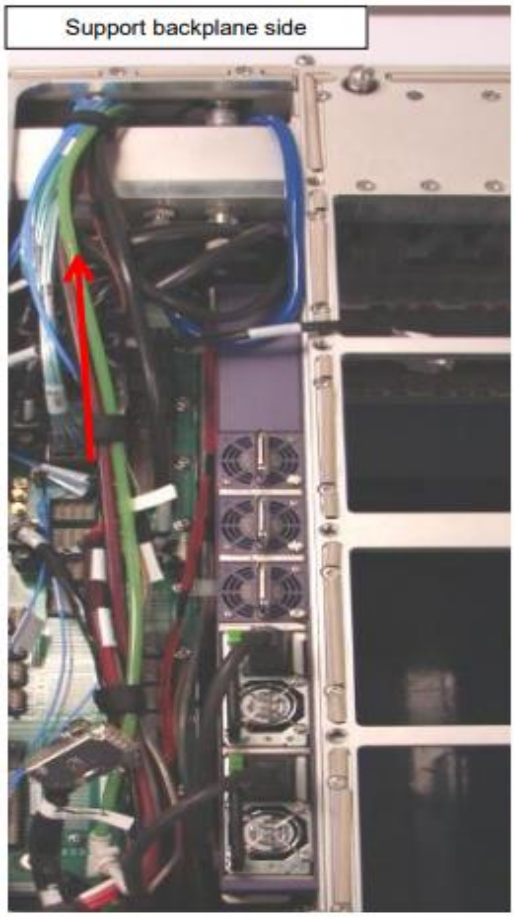
MACHINE:

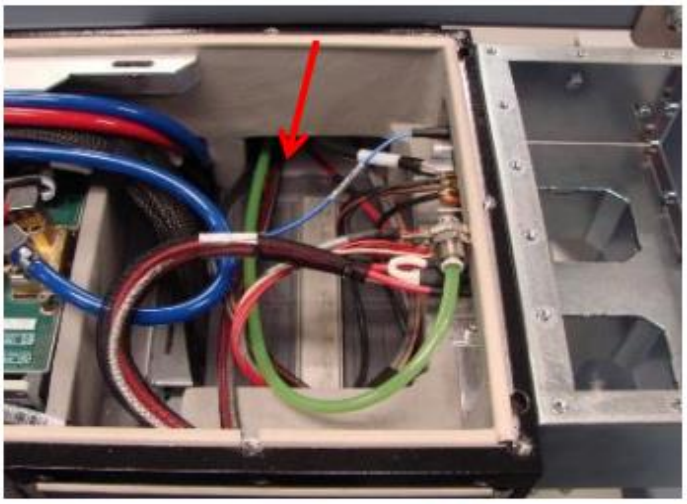
N/A

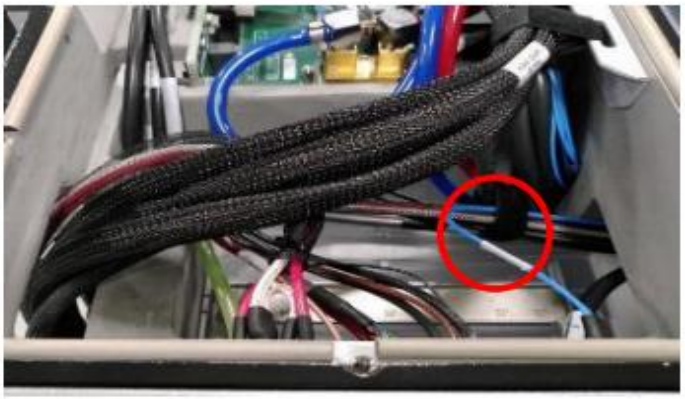
METHOD:

6.1 Put the green trachea around the Manifold block, penetrate the channel in the T/H frame and place it on the plastic cover. Use the existing cable ties to tighten the green trachea.

Use a cable tie (813-929-00) as shown in the figure to tie the cable to the backplane connection area.







MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

17mm socket wrench 3.3 N·m

METHOD:

6.2 Remove the nut on the front of the green air pipe as shown in the figure, and then adjust the nut on the back to the position 10mm from the front end.

6.3 Install the trachea connector to the panel as shown in the figure, and then install the nut to fix it. The tool is a 17mm socket wrench and the torque value is 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

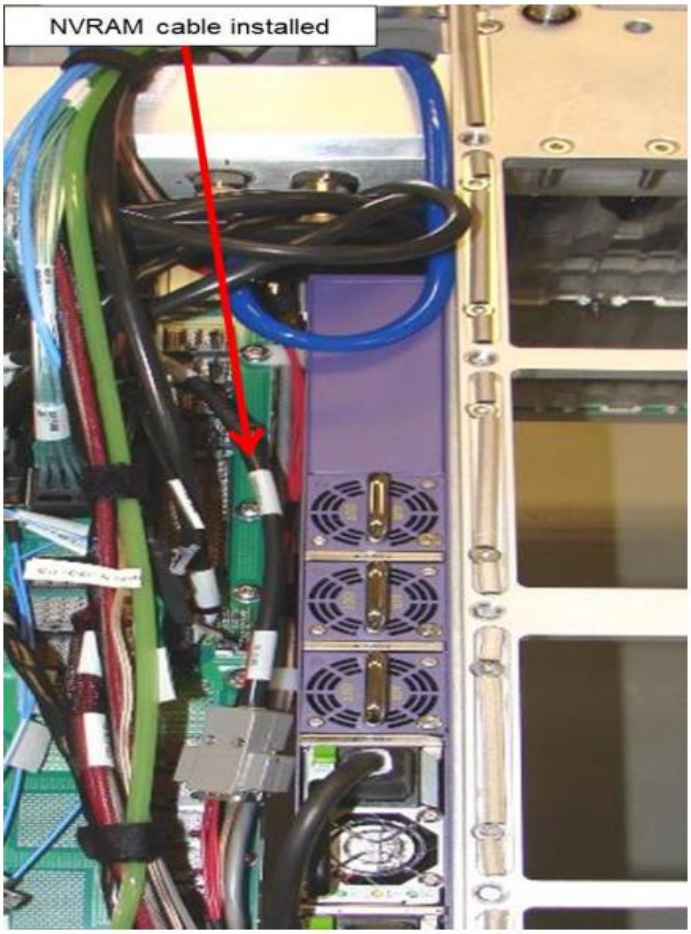
ITEM	PART NUMBER	DESCRIPTION	QTY
10	636-665-00	CABLE UF2_NVRAM_COMM 12 V W/ CONN	1

MACHINE:

N/A

METHOD:

7.1 Insert the connector of the NVRAM cable (636-665-00) labeled "Support Plane J38" into the J38 interface on the backplane PCBA as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
11	632-923-01	ALUMINUM COVER SBR BOTTOM PLATE BRACKET OR MISC	1
12	360-551-05	SHIELD EMI GASKET, D-SHAPE, .197"HX.256"W	1

MACHINE:

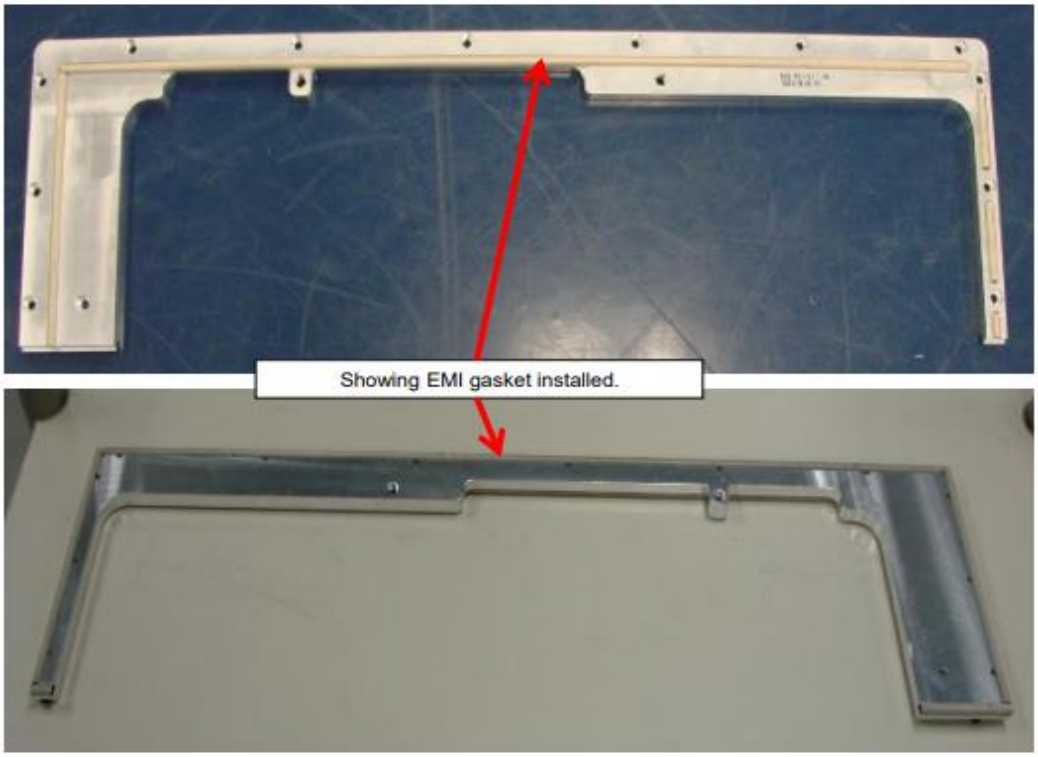
N/A

METHOD:

8.1 Take one backplane cover (632-923-01).

8.2 Install EMI foam (360-551-05) in the grooves on both sides of the cover as shown in the figure.

8.3 Place the cover plate on the material cart for subsequent assembly.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
13	406-512-12	SCRW CP SKT HD M5X.8P/12 MM SS W/NYLOK	13

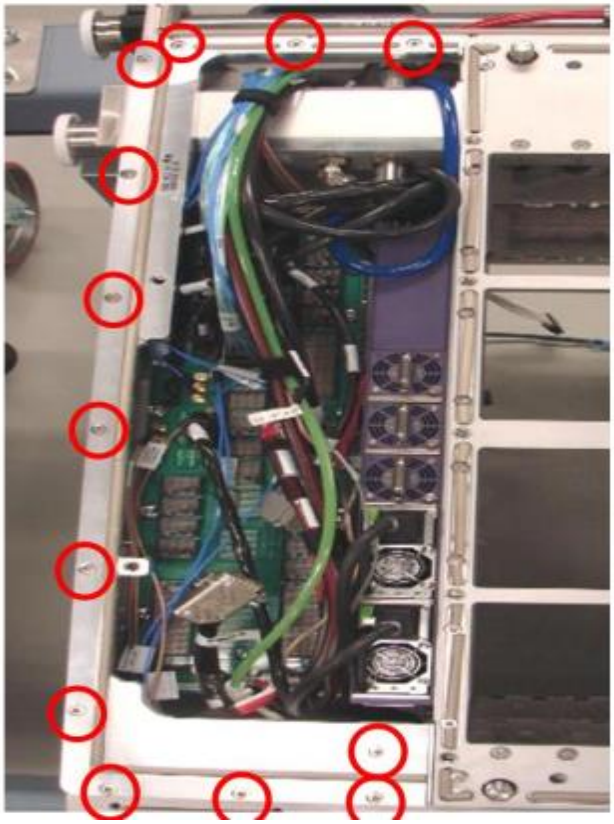
MACHINE:

4mm hex bit torque wrench 3.3 N·m

METHOD:

9.1 Install a backplane cover (632-923-01) to the T/H backplane area as shown in the figure, and use 13pcs M5 screws (406-512-12) to fix the backplane. The tool is a 4mm hex bit torque wrench with a torque value of 3.3 N·m.

9.2 All heavy gauge cables (Black cables), should be routed underneath the signal cables as shown.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
14	813-929-01	HDW VELCRO STRP 5/8	3

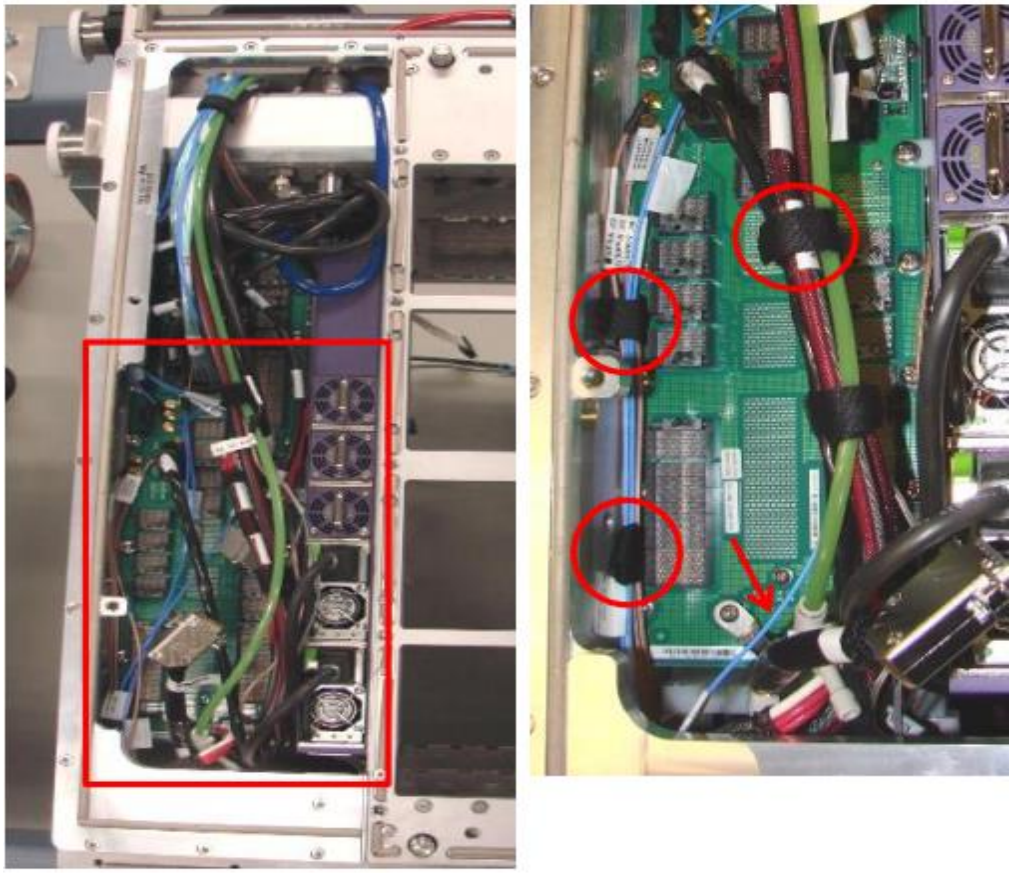
MACHINE:

N/A

METHOD:

9.3 Place the wire along the T/H frame in the area of the back plate as shown in the figure and tie it tightly with 2pcs cable ties (813-929-01). The wiring method is shown in the figure.

9.4 As shown in the figure, tie the TDR line J49 and the existing wiring harness together r with a cable tie (813-929-01).



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
15	653-411-00	PLASTIC LABEL 1-11 SLOT LABEL	1
16	653-411-01	PLASTIC LABEL 0-10 SLOT LABEL	1
17	653-411-02	PLASTIC LABEL 12-22 SLOT LABEL	1
18	653-411-03	PLASTIC LABEL 13-23 LABEL	1
19	640-211-00	PLASTIC LABEL 13-23 LABEL	1
20	656-411-00	PLASTIC, UF2 TH SBR LABEL, LABEL	1

MACHINE:

N/A

METHOD:

10.1 Take the following labels:

a. 1 UF2 TH SLOT 1-11 label (653-411-00).

b. 1 UF2 TH SLOT 0-10 label (653-411-01).

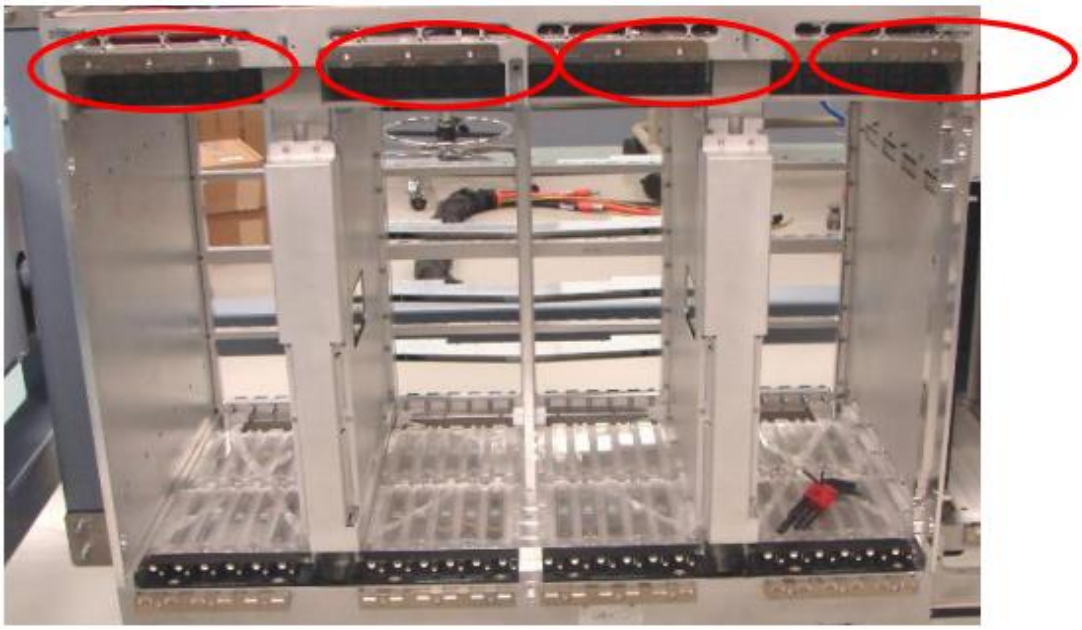
c. 1 UF2 TH SLOT 12-22 label (653-411-02).

d. 1 UF2 TH SLOT 13-23 label (653-411-03).

e. 1 Ethernet switch label (640-211-00).

f. 1 UF2 TH SLOT 22,68 label (656-411-00).

10.2 Rotate the T/H direction as shown. Note that the red circle is the location of the label.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

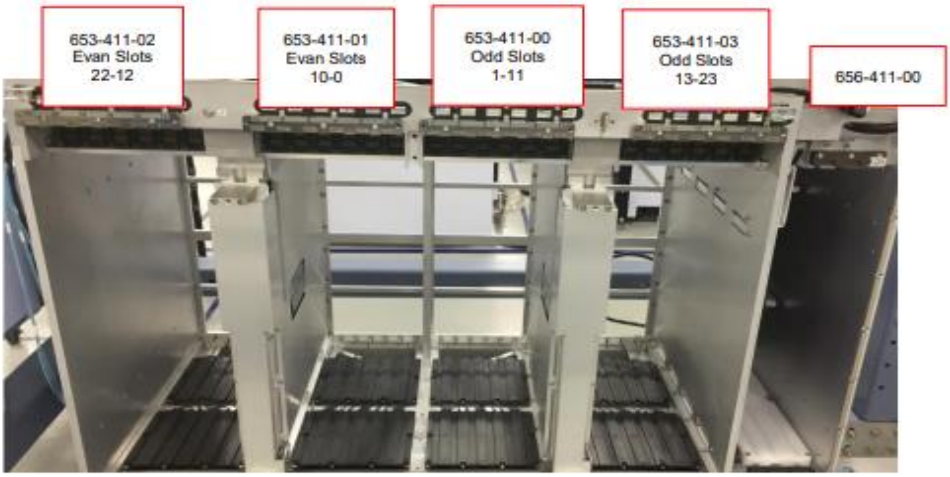
N/A

MACHINE:

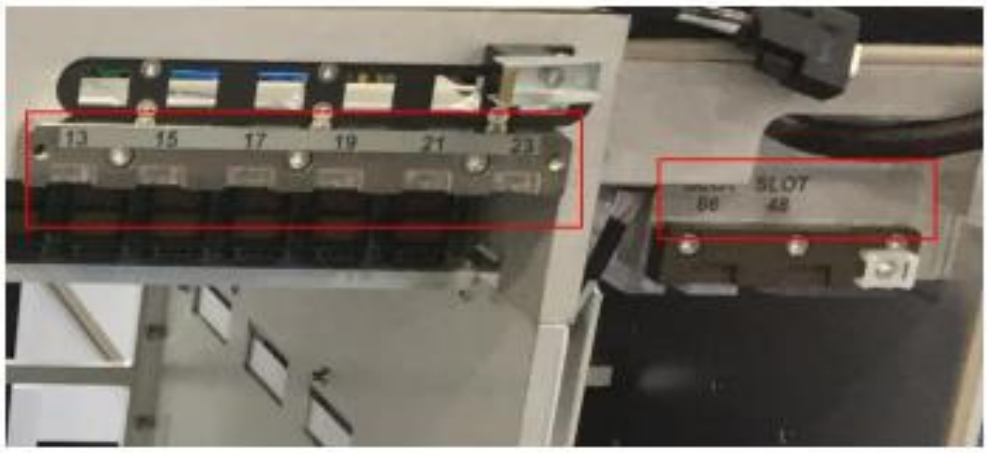
N/A

METHOD:

10.3 Install the label on the T/H frame as shown in the figure.



10.4 Install the label position as shown in the figure, the label paper is in the center, and the top position of the label is flush with the metal edge. Repeat the installation of tags in the other 3 locations.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

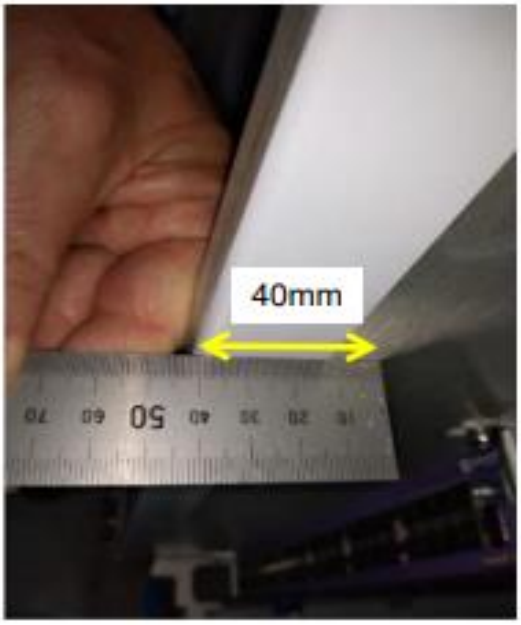
MACHINE:

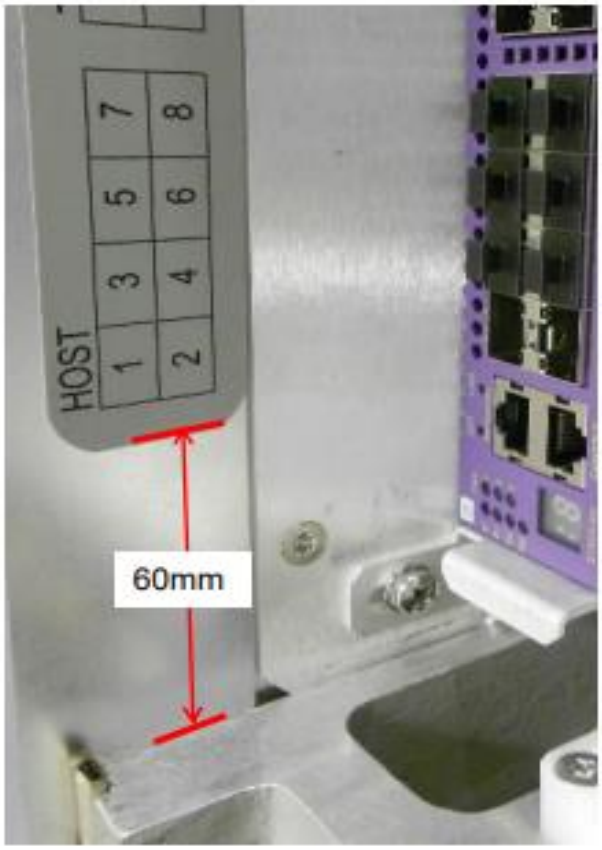
Ruler

METHOD:

10.5 Take an Ethernet switch label (640-211-00).

10.6 Install the label as shown in the picture. The edge of the label is 40mm from the left edge of the frame and 60mm from the bottom.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

N/A

MACHINE:

N/A

METHOD:

10.7 Final label assembly renderings.

Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
21	638-517-04	PLASTIC CDU COOLANT RETURN LABEL	1
22	638-517-03	PLASTIC CDU COOLANT SUPPLY LABEL	1
23	635-536-00	CABLE UF2 SB24 AOC 1M 10 V TELECOM WITH CONN	2
24	710-048-10	TRANCEIVERS PIN SFP+ MULTI 10GBS HOT PLUG DUPLEX CN	4

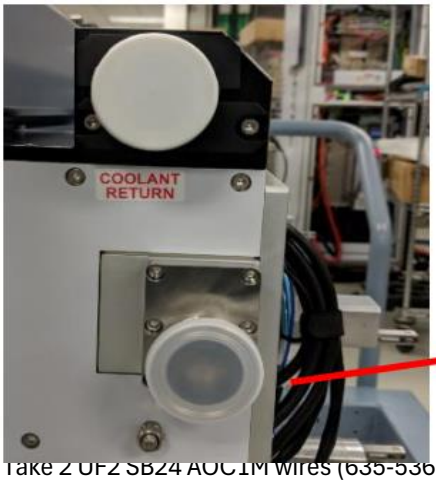
MACHINE:

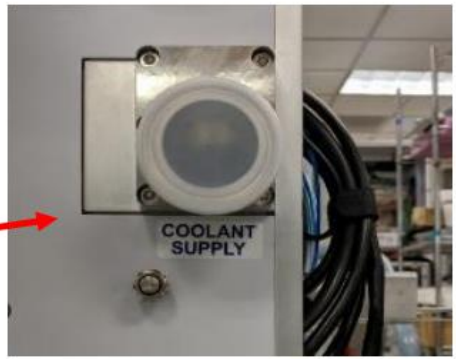
N/A

METHOD:

11.1 Put a Coolant Return label (638-517-04) on the right side of the side nozzle as shown in the figure.

11.2 Put a Coolant Supply label (638-517-03) on the right side of the inner nozzle as shown in the figure.





12.1 Take 2 UF2 SB24 AOC1M wires (635-536-00).

12.2 Take 4 transceivers (710-048-10).

12.3 Save these materials, do not open the package. This will be used in the assembly of the cabinet and T/H.

Safety Tips: N/A